

About this exercise

Thank you for your interest in joining our team as an Architect within our AI-First Engineering team. As part of our interview process, we ask candidates to complete a practical take-home exercise. Rather than testing trivia or asking you to solve puzzles under artificial pressure, this exercise is designed to reflect the actual work of the role. You'll take a business problem, research a sound design that fits within real technical constraints, and demonstrate that the design works.

We recognize that modern architecture work increasingly involves AI tools, and we embrace that. A significant part of what we're interested in is how you use AI, both to research and shape a design and to build a working prototype in an AI-powered IDE. There is no expectation that you avoid these tools. On the contrary, we want to see how effectively you put them to work.

There's nothing to submit in advance. You'll do the work on your own, and then we'll schedule a one-hour meeting where you present your solution, walk us through how you approached the problem and the assumptions you made, and demonstrate your prototype for us live.

What we're looking for

This is an open-ended exercise, and that is intentional. We will give you a business problem and enough detail about the existing solution and environment that your design must fit within, but we will not prescribe an approach. We want to see how you think.

Specifically, we're interested in:

- How you interpret an underspecified problem and frame it into something you can design against
- The soundness of your proposed design and your reasoning about tradeoffs
- Your awareness of viable alternatives and why you chose the direction you did
- How you use AI to research options and accelerate your work
- Your ability to turn a design into a working prototype that demonstrates and validates the idea
- How clearly you can communicate all of this to both technical and non-technical audiences

Because the problem is open-ended, you will need to make assumptions and decisions about scope and direction. That is expected and welcomed. There is no single "right" answer. We are as interested in the assumptions you make and the reasoning behind them as we are in the end result, so please capture them as you go.

Time expectations

We've designed this exercise to target roughly 5 to 6 hours of effort spread over about 2 to 3 days. That said, we know candidates are busy and have other commitments, so this is flexible and ultimately up to you based on your available time.

We'll ask you to tell us roughly how much time you were able to invest. Spending less time is not a negative. We assess your results in the context of the time you had available.

The prototype is meant to be a thin vertical slice that demonstrates the core of your idea. It does not need to be production-ready, fully tested, or feature-complete. Simulate or stub out any data, services, or activity that isn't central to showing your design working.

What to prepare

Come to the meeting ready to present three things. There's nothing to send us beforehand.

1. An executive summary of your proposed solution. A concise summary of the approach you selected and why. This should include the alternatives you considered but did not select, and a brief explanation of the tradeoffs that led you to your recommendation. Be ready to speak to the key assumptions and scope decisions you made along the way. Pitch it for an audience that includes both technical stakeholders and executives. Clarity matters more than length.

2. A working prototype. A functional prototype, built in the AI-powered IDE of your choice, that demonstrates your proposed design. It should simulate any required data or activity and then deliver the new capability using your selected approach. You'll demonstrate it live during the meeting. Again, this is a demonstration and validation of the idea, a thin slice rather than a finished product.

3. The story of your approach and journey. Be ready to walk us through how you tackled the exercise, which AI tools or services you used, how you approached your research, and the key decisions you made along the way. We are not asking for a prompt-by-prompt log or a detailed audit of everything you typed. A clear narrative of your journey and how AI factored into it is exactly what we're after.

The meeting

We'll schedule a one-hour live meeting where you present your work. There's nothing to send us in advance. It's a collaborative conversation, not an interrogation. We'll ask you to present your executive summary, walk us through how you approached the problem and the assumptions you made (including how you used AI along the way), and then demonstrate your prototype. It's an opportunity for us to understand your thinking and for you to show us how you reason about design.

How we'll evaluate your work

We evaluate holistically, weighted toward your design thinking and the reasoning behind it, with the prototype serving as evidence that your idea holds together. Broadly, we're assessing:

- **Design and architecture:** the soundness of your solution, your use of proven patterns and established practices, and how well it fits the given constraints
 - **Tradeoff reasoning and alternatives:** the depth and honesty of your thinking about other viable approaches
 - **AI-enabled research and build:** how effectively you used AI to research your design and to produce a working prototype
 - **Prototype quality:** how well the prototype demonstrates and validates your design
 - **Communication:** how clearly you convey your solution to both technical and non-technical audiences
 - **Handling ambiguity:** the quality of the assumptions and scope decisions you made
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Project scope

Background. Nymbus operates a multi-tenant payment processing platform that authorizes real-time debit card transactions. External payment networks connect to the platform and send ISO 8583 messages over persistent TCP connections, and the platform makes and responds with a decision on each one, typically in under 500 milliseconds. The platform is multi-tenant, meaning a single running system serves multiple bank clients at once, and each incoming transaction is routed to its owning tenant based on the card's issuing bank number (BIN).

Today, operations teams have very little real-time insight into the traffic flowing through the platform. The current metrics are little more than point-in-time counters. There is no time-series view, no way to watch trends as they develop, and no ability to slice the traffic by the attributes that matter when something goes wrong or latency starts to increase.

The business problem. We want a near real-time operational dashboard inside the existing product's operational console that gives a rolling 24-hour view of activity across the platform, refreshing on the order of a few seconds behind live. Operators should be able to filter and drill down across the meaningful attributes of the traffic to understand what is happening and investigate issues as they occur.

Because the platform is multi-tenant, the dashboard has two distinct audiences:

- Each tenant's own admin and operations team, who should be able to see the activity for their institution.
- Nymbus's own operations team acting as the platform (global) operator, who oversee the platform as a whole.

Part of the exercise is designing how the solution captures and serves data appropriately for both audiences.

What flows through the system (domain context). Each transaction is an ISO 8583 message. The messages fall into several message types (for example authorization requests, reversals, advices, and network-management or heartbeat messages), and the financial ones fall into transaction families such as purchase, withdrawal, deposit, transfer, and payment. Every transaction also produces a numeric outcome code indicating how it resolved (for example, approved, insufficient funds, exceeds limit, or various system and format conditions). Transactions carry additional attributes such as the source system they arrived from, the amount, and end-to-end processing latency. The ISO 8583 specification is the authoritative reference for these message types, data elements, and response and outcome codes, and is a good starting point for understanding what the traffic looks like.

The platform includes a connector component that interfaces with the external EFT Vendor systems, which are the sources of the ISO 8583 traffic. It receives their requests, drives the processing, and returns the responses. Because every transaction passes through it, the connector is a natural instrumentation point at which per-transaction metrics data can be captured. Defining what to capture there, and how to model and aggregate it, is part of your design.

The environment your design must fit within is not greenfield. Your design must work inside the existing platform and its runtime.

- **Language and runtime:** TypeScript on Node.js. New backend work lives in the existing services, and the frontend is Next.js and React.
- **Data and infrastructure already deployed and available to you:** PostgreSQL (AWS Aurora), a Redis or Valkey cluster (ElastiCache, already used for other shared state), Kafka (MSK), and an OpenTelemetry collector feeding Grafana for long-term observability.
- **Deployment and runtime:** the platform runs as multiple container instances in AWS EKS and scales horizontally. Instances restart during deploys, scaling, and crashes.
- **Constraints:**
 - The authorization path is latency-sensitive (sub-500 ms decisions). Metrics capture must not slow it down or create back-pressure, and if the metrics subsystem fails, transaction processing must continue unaffected.
 - Because this is banking traffic, transaction data must not be sent to any external or third-party SaaS for the real-time view.
- **Scope of the real-time system:** the target is the rolling 24-hour view within the product's existing operations console. Long-term historical observability beyond 24 hours can integrate into the existing OpenTelemetry capability and render in the existing observability platform.

Volume to design for. Up to about 50 client as environment tenants that have 1 of 5 supported EFT Vendors sending ISO 8583 traffic. Each EFT Vendor averages sending about 2 transactions per second for each client bank they serve and the most active clients receive about 15 transactions per second, so typical peak throughput is on the order of 100-150 transactions per

second across all sources on the environment. Total daily volume is roughly 2 to 3 million transactions.

What the prototype should demonstrate. A thin vertical slice that simulates representative transaction traffic, captures metrics from it using your proposed approach, and drives a near real-time dashboard that shows the traffic and lets a user filter and drill down. You do not need to implement real ISO 8583 parsing or the real pipeline. Simulate the traffic at whatever fidelity best demonstrates your design.